Edge-Compute Pacing & Feedback Node

CPSC 498 / ECE Senior Project — Concept Summary

Students: Sean Balbale & Zoe Davidow
Program: B.S. ECE / B.S. CS, Trinity College (Class of 2027)
Proposed Advisor: Prof. Kenneth A. Kousen
Date: April 28, 2026

Abstract

This project proposes the development of an embedded edge-compute node designed to provide real-time biomechanical and fatigue feedback to endurance athletes — specifically rowers and cyclists. The device interfaces with existing wireless telemetry (BLE / ANT+) to aggregate live sensor data from power meters and heart-rate monitors, executes a physiologically grounded fatigue-prediction algorithm (W' balance) directly on a resource-constrained microcontroller, and translates the result into an intuitive visual interface via a programmable LED array. The core research contribution is architectural: demonstrating rigorous, low-latency numerical computation on the edge — without offloading to a smartphone or cloud — under strict power and memory budgets.

Problem Statement

Commercial cycling and rowing head units display raw telemetry (watts, cadence, split) but perform no predictive computation. Athletes must self-interpret numbers while under physiological stress, a cognitively expensive task that degrades pacing decisions.

Smartphones and cloud offload solve this but introduce *connectivity dependency*, *latency*, and *battery vulnerability* — unacceptable in competitive race settings. The problem this project addresses is therefore:

> *Can a real-time fatigue model be implemented with sub-100 ms latency on a bare-metal or RTOS microcontroller, with sufficient numerical precision to provide actionable pacing feedback — entirely offline?*

Core Algorithm: The W' Balance Model

Physiological Background

The W' (W-prime) model, formalised by Monod & Scherrer (1965) and extended by Skiba et al. (2012), describes an athlete's finite anaerobic work capacity above their *Critical Power* (CP) threshold. Two athlete-specific parameters govern the model:

- CP — the highest sustainable steady-state power output (watts), analogous to lactate threshold.
- W' — the total finite energy reservoir (joules) available for supra-threshold efforts.

Continuous Depletion (Monod–Scherrer)

When power output $P(t) > CP$, the reservoir depletes. The instantaneous balance is:

$$W'_{\text{bal}}(t) \;=\; W' \;-\; \int_0^t \big[P(\tau) - CP\big]^+ \, d\tau \tag{1}$$

where $[x]^+ = \max(x,\, 0)$ is the positive-part operator, ensuring only supra-threshold power draws from the reservoir.

Reconstitution Model (Skiba et al., 2012)

When $P(t) < CP$, the reservoir recovers exponentially. Skiba's differential formulation unifies both phases in a single ODE:

$$\frac{dW'_{\text{bal}}}{dt} \;=\; \begin{cases} -\big(P(t) - CP\big) & \text{if } P(t) \geq CP \quad \text{(depletion)} \\ \dfrac{W' - W'_{\text{bal}}(t)}{\tau_{W'}} & \text{if } P(t) < CP \quad \text{(recovery)} \end{cases} \tag{2}$$

The recovery time constant $\tau_{W'}$ is itself power-dependent:

$$\tau_{W'}(t) \;=\; 546\, e^{-0.01\,[CP - P(t)]} \;+\; 316 \quad \text{[seconds]} \tag{3}$$

This yields fast recovery near CP and slow recovery at very low intensities, consistent with observed exercise physiology.

Discrete-Time Implementation (On-Device)

The MCU samples sensors at fixed intervals Δt (target: $\Delta t = 1\,\text{s}$ from the ANT+ broadcast). We replace the continuous integral with a first-order Euler update:

$$W'_{\text{bal}}[n] \;=\; \begin{cases} W'_{\text{bal}}[n-1] \;-\; \big(P[n] - CP\big)\,\Delta t & P[n] \geq CP \\ W'_{\text{bal}}[n-1] \;+\; \dfrac{W' - W'_{\text{bal}}[n-1]}{\tau_{W'}[n]}\,\Delta t & P[n] < CP \end{cases} \tag{4}$$

with boundary condition $W'_{\text{bal}}[0] = W'$ (athlete fully recovered at start). This requires only fixed-point arithmetic per tick — well within the Cortex-M4F budget at 64 MHz.

LED Feedback Mapping

The scalar output $W'_{\text{bal}}[n] \in [0,\, W']$ is normalised and mapped to an N-pixel LED strip:

$$k_{\text{lit}}[n] \;=\; \left\lfloor N \cdot \frac{W'_{\text{bal}}[n]}{W'} \right\rfloor, \qquad \text{hue}[n] = \begin{cases} \text{green} & k_{\text{lit}}/N > 0.50 \\ \text{amber} & 0.20 < k_{\text{lit}}/N \leq 0.50 \\ \text{red} & k_{\text{lit}}/N \leq 0.20 \end{cases} \tag{5}$$

A haptic / buzzer interrupt fires when $W'_{\text{bal}}[n] < 0.10 \cdot W'$.

Technical Architecture

Hardware Layer

- **MCU:** Nordic nRF52840 — native dual-mode BLE 5 + ANT+ radio, ARM Cortex-M4F @ 64 MHz, 256 kB RAM, 1 MB Flash.
- **Power:** 18650 Li-Ion cell + BMS; target runtime ≥ 6 h.
- **Output:** WS2812B NeoPixel strip (8–12 LEDs) + piezo buzzer.
- **Form factor:** Handlebar mount (cycling) or hull-mount (rowing shell).

Firmware & Software Layer

- **RTOS:** FreeRTOS — sensor task, compute task, output task with defined priorities and circular-buffer IPC.
- **Protocols:** ANT+ Bicycle Power Profile (PWR) and Heart Rate Profile (HRM); BLE GATT power service as fallback.
- **Language:** C / C++; algorithm unit-tested against reference Python implementation before deployment.
- **Latency target:** sensor-to-LED update ≤ 100 ms end-to-end.

Computer Science Research Framing

The hardware is a *constraint*, not the contribution. The open CS research questions are:

1. **Algorithmic correctness under discretisation.** How does Euler-step sampling frequency affect W'_{bal} accuracy relative to the continuous-time ground truth in Eq. (1)?
2. **Concurrent firmware architecture.** What FreeRTOS scheduling policy minimises worst-case latency while preventing sensor-buffer overflow on a single-core MCU?
3. **Extensibility toward ML.** Can a lightweight model (e.g., tiny LSTM) trained on historical power traces improve personalisation of $\tau_{W'}$ beyond the population-level constants in Eq. (3)?

These questions sit at the intersection of *computational methods*, *concurrent systems design*, and *AI applied to software* — directly aligned with Prof. Kousen's published research interests.

Deliverables & Proposed Timeline

Phase	Milestone	Target
1	Hardware bring-up; BLE / ANT+ sensor pairing verified	Sep 2026
2	W' algorithm on MCU; validated against Python reference	Oct 2026
3	RTOS integration; latency and scheduling benchmarks	Nov 2026
4	Full feedback loop; field test with live power meter	Dec 2026
5	Formal write-up, poster, and final public demo	Apr 2027

What We Are Asking of an Advisor

- **Check-ins** to review software architecture and algorithmic design decisions.
- **Feedback on written deliverables** — proposal, interim report, and final thesis — well-suited to Prof. Kousen's background as a published technical author.
- **Guidance on CS rigour:** framing the project's contribution to satisfy CPSC 498 standards, particularly on the algorithmic and systems-design components.
- **Optional:** advising on any ML-based extensions (personalised $\tau_{W'}$ estimation) pursued in Phase 5.

Deep embedded-hardware expertise is **not** required — hardware selection is settled. The open problems are software architecture and algorithmic design.

Contact: Sean Balbale — sean.balbale@trincoll.edu | Zoe Davidow — zoe.davidow@trincoll.edu